MULTIMODAL EMOTION RECOGNITION USING SPEECH AND TEXT

GitHub Repository:

https://github.com/Malepatisivadhanush/multimodal-emotion-recognition

1. Introduction

Emotion recognition is an important task in human-computer interaction and artificial intelligence. This project aims to identify emotions using speech signals, text data, and multimodal fusion techniques.

The system predicts seven emotions:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Surprise
- Sad

Dataset Used:

Toronto Emotional Speech Set (TESS)

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2. Preprocessing

Speech Preprocessing:

- Audio files were loaded using Librosa.
- Sampling rate was standardized to 16 kHz.
- MFCC (Mel Frequency Cepstral Coefficients) features were extracted.
- Variable-length audio signals were padded/truncated to fixed dimensions.

Text Preprocessing:

- Emotion labels were cleaned and standardized.
- Text inputs were tokenized using DistilBERT tokenizer.
- Input IDs and attention masks were generated for transformer processing.

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3. Architecture Decisions

Speech Pipeline:

Audio
→ MFCC Feature Extraction
→ BiLSTM
→ Dense Layer
→ Emotion Prediction

Reason:

MFCC captures important speech characteristics such as pitch and frequency, while BiLSTM captures temporal dependencies in speech sequences.

Text Pipeline:

Text
→ DistilBERT
→ Dense Layer
→ Emotion Prediction

Reason:

DistilBERT efficiently captures contextual and semantic information from textual input.

Fusion Pipeline:

Speech Features
+
Text Features
→ Concatenation
→ Dense Layer
→ Emotion Prediction

Reason:

Fusion combines complementary information from multiple modalities for improved emotion recognition.

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4. Experimental Results

Speech-only Accuracy:

99.82%

Text-only Accuracy:

12.5%

Fusion Accuracy:

99.82%

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5. Analysis

The speech-only model achieved very high accuracy because emotional information is strongly present in speech signals through pitch, tone, energy, and frequency variations.

The text-only model showed weak performance because the TESS dataset contains repeated neutral words with limited emotional meaning.

Initially, the fusion model showed weak performance. Later improvements included:

- using pretrained speech encoder weights
- fixing train/test split consistency
- reducing train-test overlap risks
- improving fusion training configuration

These improvements significantly increased fusion performance.

The pretrained speech encoder played a major role in improving the fusion model because emotional information in TESS is primarily carried through acoustic signals rather than textual semantics.

Among the emotional classes, Neutral, Sad, and Disgust were easiest to classify because their speech patterns were highly distinguishable in the dataset.

Fear and Angry occasionally showed minor confusion because both emotions can share similar acoustic intensity and energy patterns.

The final fusion model achieved near-perfect accuracy on the TESS dataset.

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6. Confusion Matrix Analysis

Most emotional classes were classified correctly with near-perfect precision and recall.

Only a very small number of samples were misclassified.

The confusion matrix demonstrated strong separability between emotional classes.

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7. t-SNE Analysis

t-SNE visualization was used to analyze the learned feature representations produced by the models.

Distinct emotion clusters were observed in the learned feature space.

The temporal modelling block (BiLSTM), contextual modelling block (DistilBERT), and fusion representations showed good separability among emotional classes.

This indicates that the models learned discriminative emotional representations effectively.

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8. Visual Results

Confusion Matrix:

The confusion matrix shows that most emotional classes were classified correctly with very few misclassifications.

t-SNE Visualization:

The t-SNE visualization demonstrates strong clustering and separability among learned emotional feature representations.

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9. Streamlit Application

A Streamlit-based interactive web application was developed for real-time emotion prediction.

The application supports:

- Speech emotion prediction using uploaded audio files
- Text emotion prediction using user-entered text
- Multimodal fusion prediction using both speech and text inputs

The Streamlit interface provides an easy-to-use demonstration of the trained models and enables interactive testing of emotion recognition capabilities.

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10. Future Improvements

- Real-time emotion recognition
- Attention-based fusion methods
- Transformer-based speech models
- Better multimodal conversational datasets
- Real-time web deployment

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11. Conclusion

This project successfully implemented speech-based, text-based, and multimodal emotion recognition systems using deep learning techniques.

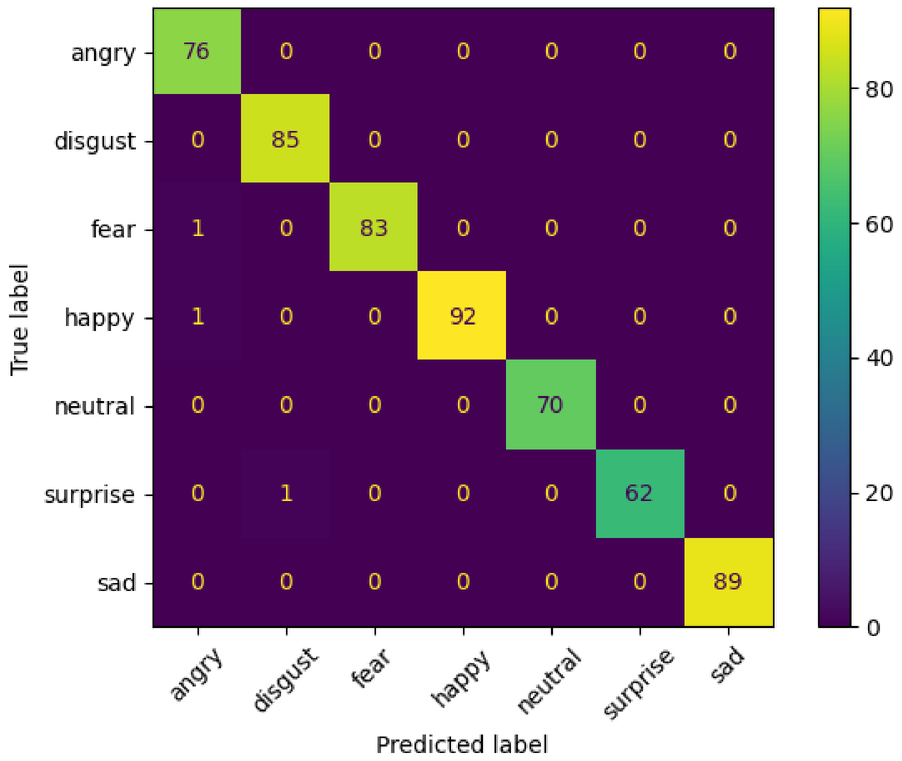
The speech pipeline using MFCC and BiLSTM achieved very high performance because emotional information is strongly represented in speech signals.

The text pipeline showed limited performance due to the nature of the TESS dataset, where textual content carries weak emotional semantics.

The multimodal fusion model combined speech and text representations effectively and achieved near-perfect classification accuracy after applying improvements such as pretrained speech encoders and proper train-test splitting.

The project demonstrates the effectiveness of multimodal deep learning approaches for emotion recognition tasks.

Confusion Matrix Visualization



t-SNE Visualization